International Entrepreneurship and Management Journal (2026) 22:45
https://doi.org/10.1007/s11365-026-01185-x

Resilience in startups: a multidimensional and processual perspective on psychological, organizational, and financial adaptation

Fauzia Jabeen[1] · **Gabriele Santoro**[2,3] · **Tomas Kliestik**[4] · **Dorra Yahiaoui**[5]

Accepted: 17 December 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2026

Abstract
Resilience is increasingly recognized as a key factor for startup growth and survival. However, the relevant literature remains scarce. This study explored how resilience is enacted in early-stage ventures by identifying its underlying mechanisms and dynamics. A qualitative multiple case study with seven startups operating across different sectors and contexts was conducted. We obtained data from 27 semi-structured interviews with founders and managers, supplemented with secondary data. The findings revealed that startup resilience is not a fixed capability but a recursive process shaped by the dynamic interaction of three key dimensions: psychological or individual, organizational or strategic, and financial. Novel underlying mechanisms for each domain were identified: cognitive grounding, affective anchoring, and emotional regulation at the psychological level; experimental responsiveness, modular recomposition, and sense-giving at the organizational level; and financial reflexivity, cash flow tactics, resource reconfiguration, and temporal elasticity at the financial level. These crucial dimensions reinforced each other in a looped architecture of resilience. This study contributes to the literature by conceptualizing resilience in startups as a process of recursive recomposition driven by psychological, organizational, and financial factors. It also provides guidance for startup founders and managers on how to cope with hardship through routines, tactics, and postures.

Keywords Startups · Resilience · New ventures · Entrepreneurship

Extended author information available on the last page of the article

Published online: 20 February 2026

Introduction

Startups operate in inherently uncertain and volatile environments, facing financial constraints, competitive pressures, and market unpredictability (Sommer et al., 2009). Unlike established firms, startups lack data records, a well-established and defined business model, stable revenue streams, and institutional support. This makes them particularly vulnerable to external threats, changes, and shocks (Kuckertz et al., 2020). Their initial value propositions and business models are often modified and innovated over time to achieve product–market fit (Zott & Amit, 2008), which can be influenced or threatened by numerous internal and external factors. Therefore, new ventures must build a sustained capacity to cope with external difficulties, thus building resilience. Resilience, defined as the capacity to withstand and adapt to adversity, has emerged as a crucial determinant of startup survival and long-term success (Hillmann & Guenther, 2021) as well as entrepreneurial growth and survival (Hedner et al., 2011; Duchek, 2018; Bohnsack& Liesner, 2019). The concept of resilience originated in the natural sciences, specifically ecology, where it was used to describe an ecosystem's capacity to absorb disturbances and reorganize while undergoing change to retain its essential functions, structure, and feedback mechanisms (Holling, 1973). Subsequently, the concept was adopted in other fields such as psychology, where it was employed to understand individuals' abilities to cope with adversity (Masten, 2001). This interdisciplinary evolution has led to the adoption of resilience in organizational and management fields as a framework for understanding how organizational members cope with uncertainties and difficulties. For example, resilience is often defined as organizations' ability to anticipate, adapt to, and recover from unexpected challenges (Vogus & Sutcliffe, 2007; Conz & Magnani, 2020; Santoro et al., 2021). In this context, it has been linked to various strategic and operational factors such as agility (Doz & Kosonen, 2010), innovation (Lengnick-Hall et al., 2011), and stakeholder collaboration. In the context of entrepreneurship, resilience is regarded as the ability to cope with serious individual and organizational threats and take actions accordingly (Santos et al., 2023). Most studies have primarily focused on large, established organizations, examining their capacity to navigate economic downturns, competitive pressures, and technological disruptions (Santoro et al., 2020; Grego et al., 2024). Nevertheless, research is increasingly including startups and small businesses, highlighting the unique challenges pertaining to resilience faced by these entities owing to their limited resources and higher vulnerability (Roundy et al., 2017; Haase & Eberl, 2019; Cuéllar-Fernández et al., 2024). Hence, the ability to navigate crises, pivot business strategies, and sustain operations despite setbacks is being increasingly recognized as a key entrepreneurial competency for startups (Williams & Shepherd, 2016; Morris et al., 2022). Accordingly, startups must be able to cope with difficulties and pivot repeatedly before scaling (Flechas Chaparro & de Vasconcelos Gomes, 2021; Cavallo et al 2024). Nevertheless, startup resilience is practically and conceptually different from the resilience of large organizations; thus, scholars should consider firm size when analyzing organizational resilience.

Research on resilience spans multiple disciplines, including psychology, organizational behaviour, and strategic management (Lopez et al., 2024; Roffia & Dabić, 2024). At the individual level, psychological resilience enables entrepreneurs to per-

sist in the face of failure and uncertainty by leveraging adaptability and emotional regulation to sustain their ventures (Chadwick & Raver, 2020). Furthermore, at the organizational level, adaptive business models, agile decision-making, and strong stakeholder networks enhance resilience by enabling startups to respond dynamically to environmental changes (Brykman & King, 2021; Mota et al., 2022). Meanwhile, financial and resource resilience strengthens a startup's ability to navigate economic downturns (Roundy et al., 2017; Taghavy et al., 2024). Studies suggest that entrepreneurial resilience plays a vital role in fostering employee and organizational resilience, as the entrepreneur usually interacts with other members daily and can influence them (Santoro et al., 2020, 2021; Palmucci et al., 2024).

Despite the growing interest in resilience, research on how it manifests in startups remains fragmented. Studies often examine resilience through isolated disciplinary lenses (e.g., psychology, business strategy, and finance) but lack an integrated framework for resilience in startups. Such a framework is critical as startups face unique constraints and challenges that necessitate a context-specific understanding of resilience (Blank, 2018). Startups are inherently vulnerable to survival challenges owing to their limited resources and a lack of solid organizational structures (Zahra, 2021; Fuertes-Callén et al., 2022). For example, startups and new ventures are often associated with the founders' willpower and psychological resilience when facing periods of substantial uncertainty (Kim et al., 2023). Research highlights several factors that contribute to the greater susceptibility of startups during crises, including lower levels of preparedness, greater overall vulnerability, dependence on local government and institutional support, and aggravated psychological and financial pressures on business owners (Brykman & King, 2021). These characteristics underscore the need to examine the resilience of startups, as understanding how these firms cope with adversity is critical to their long-term sustainability and growth (Cristofaro et al 2025).

Finally, while research has made valuable contributions by exploring psychological, organizational, and financial aspects of resilience in entrepreneurial contexts, these factors have been examined in isolation. This fragmented approach limits our understanding of how startups, which face simultaneous emotional, structural, and resource-related challenges, enact resilience in practice. To address this limitation, an integrated framework is required that captures the interplay between these dimensions, reflecting the recursive and multifaceted nature of resilience in early-stage ventures.

This study attempted to bridge these gaps by exploring how resilience is enacted in early-stage ventures, identifying its underlying mechanisms and dynamics. We addressed the following research question: *How do early-stage startups enact resilience when facing adversity*, *and through what mechanisms does this resilience emerge and evolve across psychological*, *organizational*, *and financial dimensions?*

To this end, we conducted a qualitative multiple case study with seven startups operating in different sectors and contexts. Data were obtained from 27 semi-structured interviews with founders and managers, along with secondary data. Our findings revealed that startup resilience is not a fixed capability but a recursive process shaped by the dynamic interaction of psychological or individual, organizational or strategic, and financial dimensions. We identified novel mechanisms underlying

each domain: cognitive grounding, affective anchoring, and emotional regulation at the psychological level; experimental responsiveness, modular recomposition, and sense-giving at the organizational level; and financial reflexivity, cash flow tactics, resource reconfiguration, and temporal elasticity at the financial level. These dimensions reinforced each other in a looped architecture of resilience.

This study contributes to the literature on entrepreneurship and resilience by reconceptualizing startup resilience as more than a static capability or binary outcome. It views startup resilience as a process of recursive recomposition that unfolds across three interdependent dimensions: psychological, organizational, and financial. Drawing on rich qualitative evidence from seven early-stage platform-based startups, we theorized how founders and teams enact resilience through cognitive grounding, affective anchoring, experimental responsiveness, financial reflexivity, and other emergent mechanisms. By doing so, we go beyond linear and siloed interpretations of resilience—often framed as individual grit, agile pivots, or financial endurance—and develop an integrated perspective that reveals how these dimensions interact dynamically. Our framework captures how resilience is not only a reactive force in times of crisis but also a proactive capability rooted in the everyday practices, postures, and routines that shape strategic and emotional coherence under uncertainty. Practically, this study offers guidance for founders and managers of new ventures by unpacking how specific routines, tactics, and cognitive postures can help startups cope with hardship and regain traction under conditions of uncertainty. The findings demonstrate how small, low-cost interventions can collectively foster resilience without the need for resource abundance. They also suggest that successful adaptation does not rely solely on agility but also on the mutual reinforcement of emotional grounding, strategic responsiveness, and financial discipline.

The remainder of the paper is structured as follows. Section 2 presents the theoretical background by discussing studies on resilience, entrepreneurship, and startups along with the main theories used in our study, such as dynamic capabilities. Section 3 explains the methodological steps of our qualitative study. Section 4 presents the qualitative results. Section 5 outlines the major theoretical and practical implications of the study.

Theoretical background

The concept of resilience has gained attention in entrepreneurship research as startups face mounting pressures from uncertain markets, technological turbulence, and unforeseen disruptions (Corvello et al 2023; Santos et al., 2023). Unlike established firms, startups lack operational buffers, institutional legitimacy, and resource redundancy, making them particularly vulnerable to adverse events (Blank, 2018; Morris et al., 2020; Zahra, 2021). Within this context, resilience has emerged as a core capability for navigating volatility, adapting to change, and sustaining entrepreneurial momentum. Nevertheless, despite a growing interest in the topic, the literature on startup resilience remains scarce and fragmented.

From a psychological and individual perspective, resilience can be understood as individuals' capacity to stay motivated, regulate emotions, and recover from setbacks

(Masten, 2001; Chadwick & Raver, 2020). For entrepreneurs, this entails the ability to cope with failure and handle high-stress situations (Bullough & Renko, 2013; Ayala & Manzano, 2014; Santoro et al., 2020). This psychological strength is not merely innate but can be cultivated and reinforced by experience, self-efficacy, self-awareness, and interpersonal interactions. In this regard, two theoretical lenses provide a rich foundation for interpreting entrepreneurial resilience. First, positive psychological capital comprises four psychological resources (hope, efficacy, resilience, and optimism) as crucial predictors of performance and well-being in high-pressure environments (Luthans et al., 2007). Positive psychological capital positively influences individual persistence and confidence as well as the ability to mobilize others under uncertainty, making it especially relevant in startup contexts. Second, post-traumatic growth (Tedeschi & Calhoun, 2004) suggests that individuals may experience considerable personal development after adverse or traumatic experiences. For startup founders, financial losses or market failures may represent catalysts for cognitive reframing, deeper self-understanding, and renewed strategic clarity.

From an organizational perspective, resilience is discussed through the lens of strategic flexibility and adaptability (Castelló-Sirvent et al., 2024). Dynamic capabilities (Teece et al., 1997; Corvello et al., 2024) provide a useful explanatory framework and refer to a firm's ability to identify opportunities and threats, seize new opportunities, and reconfigure internal and external assets to maintain competitiveness. For startups, which are embedded in turbulent environments, these capabilities are critical for iteratively adapting products, business models, and strategic goals. This perspective aligns closely with empirical evidence that suggests that resilient startups can pivot effectively, repurpose existing assets, and rapidly respond to market feedback (Silva et al., 2020; Flechas Chaparro & de Vasconcelos Gomes, 2021; Rehman et al., 2025). Ambidexterity provides another useful lens (O'Reilly & Tushman, 2013) by capturing the dual capacity to exploit existing knowledge while exploring novel domains. Exploitation concerns immediate operational efficiency, while exploration enables innovation and adaptation. According to this perspective, resilient startups must balance these competing logics. For instance, they might scale a validated core offering while experimenting with adjacent market solutions (Sanasi, 2023). Finally, lean startup approaches (Ries, 2011) provide a methodological approach to organizational resilience. Rooted in principles of iterative testing, validated learning, and customer-centric experimentation, the lean startup methodology allows founders to minimize resource waste while maximizing adaptability. The process involves building a minimum viable product, exploring feedback loops, and continuous experimentation. Startups practicing lean methods are better positioned to identify errors early, adjust course, and integrate resilience into their operational DNA. Scholars have linked lean approaches to increased strategic responsiveness and survival rates in early-stage ventures (Silva et al., 2020; Bargoni et al., 2024a, b; Macca et al., 2025).

Therefore, it is reasonable to infer that startup resilience represents the interaction of psychological strength, strategic flexibility, and operational adaptability. A psychologically resilient founder is more likely to foster a culture of learning and safety, which supports agile decision-making and innovation (Branca et al., 2025). Organizational structures that promote dynamic capabilities and ambidexterity enable flexible resource allocation and strategic repositioning (Teece et al., 2016). Furthermore,

lean methodologies operationalize these capacities, transforming resilience from an abstract concept into a set of repeatable behaviors and routines.

In this study, we drew upon these theoretical insights to interpret how resilience is enacted in practice. By analyzing qualitative cases of startups navigating adversity, we sought to uncover the context-specific mechanisms through which these capabilities emerge and interact. Our aim was not merely to describe resilience but to understand how it is built, sustained, and leveraged by startups operating under conditions of uncertainty.

Methodology

Research design

The limited research on resilience in startups and its intersection with behavioral, strategic, and financial dimensions makes the case study method appropriate for comprehensively investigating this phenomenon. Moreover, our research question, centered on understanding how resilience is developed and enacted across different startup contexts, necessitated a qualitative approach. As such, this study adopted a multiple case study design to explore how startups respond to adversity and build resilience over time (Eisenhardt & Graebner, 2007). The multiple case method allows for the identification of patterns and mechanisms across different organizational contexts, balancing analytical generalizability with deep, context-specific insights (Eisenhardt, 1989; Yin, 2018).

Case selection

We employed purposive sampling to select startups that had experienced significant adversity such as market shocks, funding gaps, operational breakdowns, and external disruptions and had undertaken strategic, organizational, or psychological responses to cope and adapt. Each selected startup also exhibited clear entrepreneurial activity and dynamic growth trajectories, often characterized by experimentation, iteration, and strategic pivoting. In total, seven startups were selected, each representing different sectors and stages of development. The startups were anonymized and assigned pseudonyms as follows:

- Batman (Travel and Tourism)
- Bane (Fintech)
- Joker (Sports and Wellness)
- Catwoman (Food Delivery)
- Nightwing (Fitness App)
- Penguin (Artificial Intelligence (AI)-Based Small and Medium-Sized Enterprise Support System)
- Riddler (AI for Pharmacy Supply Chains)

These companies varied in funding stage, industry domain, organizational size, and maturity. Batman and Bane were in the Series B funding rounds; Joker and Catwoman were in Series A; Nightwing, Penguin, and Riddler were at the earlier stages, with bootstrapped or seed funding. All firms demonstrated a proactive approach to dealing with uncertainty, making them suitable cases for examining startup resilience.

The selected startups spanned several industries and business models, each reflecting a distinct approach to value creation and resilience-building in highly dynamic environments. Table 1 presents the case studies.

These models were quite diverse, ranging from subscription services to transactional platforms to AI-based Software as a Service (SaaS) solutions. Therefore, these case studies offered a rich empirical variety for exploring how resilience mechanisms manifest under different conditions of uncertainty, resource constraints, and growth trajectories.

Data collection

The data were collected through semi-structured interviews and supplemented with secondary sources. The interviews were conducted with founders, growth managers, marketing managers, and other key actors who had directly experienced adversity or played a role in the company's response to adversity. In total, 27 interviews were conducted across the seven cases, each lasting between 60 and 90 min. For Batman, we conducted 10 interviews over a four-year period with four participants. For Bane, six interviews were conducted with the founder and growth manager. Joker and Catwoman each contributed two interviews. For the newly added cases, Nightwing (fitness) contributed three interviews (with the founder, product lead, and growth advisor); Penguin (AI for SMEs) provided one extended interview with the CEO and one follow-up interview with the AI team lead; and Riddler (pharmacy supply chains) contributed two interviews with the cofounder and operations director.

The interviews followed a semi-structured protocol, enabling cross-case comparability and contextual flexibility. The topics included the nature of the challenges encountered, internal and external responses to those challenges, decision-making processes, and reflections on resilience. All interviews were recorded with consent, transcribed verbatim, and coded for analysis. To enrich and triangulate our data, we also collected data from secondary sources such as pitch decks, investor reports, company websites, blog posts, press releases, and relevant media coverage.

We adopted a saturation logic to determine the sufficiency of the number of interviews. Data collection was concluded when new interviews no longer yielded substantially new insights or themes and when key mechanisms began to recur across cases. This theoretical saturation criterion ensured adequate depth and breadth for developing a robust and grounded interpretation of startup resilience.

Data analysis

We adopted an inductive analysis strategy, grounded in the axial coding principles (Strauss & Corbin, 1998). Transcripts and textual materials were reviewed for open codes, which were iteratively grouped into higher-order categories. Coding focused

Table 1 Startups and descriptions

Startup's name	Description
Batman	It operates in the travel and tourism sector and is structured as a two-sided platform that connects experience providers (e.g., local guides and cultural institutions) with travelers seeking personalized itineraries. Its business model is transaction-based, with revenue derived from booking commissions and Business2Business (B2B) partnerships.
Bane	This is a fintech platform offering digital financial services, such as automated investment portfolios and micro-credit options for underbanked users. It employs a freemium model, with free basic features and premium tools available through monthly subscriptions. The company also generates revenue through financial product referrals and partner Application Programming Interface (APIs).
Joker	This is a sports and wellness platform that combines streaming workouts, personal coaching, and gamified challenges. It operates under a hybrid model: direct-to-consumer (via app subscriptions) and B2B (corporate wellness programs). Joker also monetizes through branded partnerships and fitness gear sales.
Catwoman	It is active in the food delivery market, with a specialization in underserved semi-urban zones, and differentiates itself by optimizing delivery logistics through AI and offering curated local menus.
Nightwing	It is a fitness-tech startup offering a mobile app integrating personalized training programs, biometric tracking, and mental coaching. Its business model blends paid app subscriptions with upselling of premium services such as one-on-one coaching, and strategic B2B collaborations with gyms and wellness professionals.
Penguin	It provides an AI-powered administrative assistant for small and medium-sized enterprises (SMEs), focusing on improving financial planning, document management, and compliance reporting. It adopts a SaaS model with tiered pricing plans based on user volume and advanced feature access.
Riddler	It offers an AI-based solution tailored to pharmacy supply chains. It uses predictive analytics to optimize stock management, reduce waste, and improve service levels in local pharmacies. The business model includes subscription fees and service agreements with pharmaceutical distributors.

on identifying mechanisms, routines, behaviors, and cognitive frames that shaped the startups' responses to adversity. The thematic structure was informed by the literature on startup resilience, particularly the dimensions of psychological or individual, organizational or strategic, and financial or resource-based resilience. The coding

process was supported by a qualitative analysis software (NVivo) and carried out in a recursive, abductive manner, iterating between data and theory.

Following Cloutier and Ravasi (2021), we structured the data in a comparative matrix, with rows representing emergent themes and columns representing individual cases. This allowed us to observe both within-case richness and cross-case patterns, facilitating a grounded understanding of how different resilience mechanisms interacted across different contexts. We particularly focused on identifying configurations, rather than linear relationships, between psychological traits (e.g., founder mindset), organizational practices (e.g., agility and pivoting), and financial choices (e.g., burn-rate strategies and fundraising behaviors).

Ensuring trustworthiness

To ensure credibility and analytical robustness, we employed several qualitative validation techniques. First, triangulation was used by integrating interviews with multiple sources of secondary data. Second, member checking was conducted with the participants to validate key interpretations and ensure factual accuracy. Third, peer debriefing among the research team allowed for critical discussions of emerging codes and thematic categories, thereby reducing individual bias. Finally, we provided detailed contextual "thick descriptions" of each case, supporting transparency and enabling readers to assess our findings' transferability to other settings.

Findings

This section presents and discusses how startups enacted resilience during periods of substantial uncertainty and disruption. We distinguished three interdependent yet analytical dimensions of resilience: psychological, organizational, and financial. While these dimensions sometimes coevolved, each displayed unique mechanisms, dynamics, and adaptive logics. Across the cases, resilience was not a latent trait, but a dynamic capability enacted through communication rituals, structural reconfigurations, and resource orchestration.

Psychological resilience: leading with vulnerability, presence, and cognitive framing

Psychological resilience was foundational across all cases. Rather than manifesting as stoic endurance or charismatic leadership, it emerged as the consistent ability of founders, managers and teams to stabilize emotions, share fears openly, and ritualize meaning in difficult times. In each case, resilience was not an individual capability but a collective emotional architecture driven by shared feelings, emotions, ideas, and fears.

At Batman, the collapse of global travels during the pandemic presented an existential threat. Nevertheless, the team responded with daily calls not to discuss KPIs but simply to stay emotionally present and try to find effective solutions. The founder recalled:

> *Every day at 9:00 a.m., we joined the Zoom room. We didn't always know what to say. Sometimes we remembered past trips. Sometimes we just sat in silence. But we showed up. That mattered. It meant that even if there's no product, there's still a team. The idea was to share ideas about the huge issue we were witnessing.*

This intentional ritual of emotional continuity became a buffer against isolation, burnout, and strategic drift. In Bane, where market volatility was relentless, psychological resilience was framed in terms of emotional self-discipline. The co-founder explained:

> *Crypto is chaos. But chaos isn't the enemy, confusion is. I told my team that our role is to be the calmest person in the room. That became a behavioral standard. We didn't fix user panic. We absorbed it, regulated it, and turned it into structured action.*

Such cognitive reframing transformed emotional shock into organizational clarity. In this context, cognitive framing referred to the process by which individuals or groups face complex, ambiguous, or adverse situations by selecting certain aspects of reality and organizing them into coherent narratives or mental models to guide action.

Joker demonstrated a relational form of resilience: when their clients (sports clubs) shut down, they began calling them to listen, not to sell. The CEO explained:

> *We talked to 60 clubs in two weeks. Not one call was about money. We asked them, "How are you? What do you need?" That changed our culture. We weren't a directory anymore; we were partners. And the team felt it too. They had a reason to keep building.*

At Catwoman, emotional fatigue followed a hypergrowth phase. To process the post-boom stagnation, the team implemented weekly "reflection rituals." The COO shared:

> *We introduced a one-hour meeting called "Friday Reflections." No KPIs. Just a space to share something we noticed, something funny, or something hard. It sounds small, but it was a lifeline. It helped people stay connected when growth slowed.*

At Nightwing, a fitness startup for young users' modular digital workouts, psychological strain emerged not from external collapse but from internal misalignment and self-doubt. After experiencing a steep drop in user retention, the founder described the emotional disorientation of seeing their vision collapse:

> *I had convinced myself that we had achieved product–market fit. When the metrics crashed, it felt like the ground disappeared. I couldn't sleep. I started questioning everything. But then I decided to stop hiding behind dashboards. I joined every single user support call for a week. I just listened. And that gave me some clarity not about the product but about why we were building it.*

This act of returning to the human side of the problem, speaking directly with users, confronting their frustration, and absorbing feedback became a ritual for regaining

psychological grounding and rebuilding strategic clarity. This underscored that, when facing difficulties, it can be beneficial to revisit the fundamentals of business and revise cognitive structures.

In Penguin, an SaaS solution for small businesses, psychological resilience manifested at the team level by reframing loss as focus. Their CEO stated:

> *When 35% of our customers churned in two months, we were devastated. But then our CTO said something brilliant, "Let's stop worrying about the users who left and start focusing on the ones who stayed." That changed everything. It gave us a target, a reason to rebuild.*

This reframing catalyzed a morale shift and instilled a new internal narrative: resilience was no longer about bouncing back; it was about emphasizing the value they could still offer.

In sum, psychological resilience was enacted through small but consistent cultural practices and routines, rituals of presence, empathy, and re-narration. This allowed teams to metabolize fear, reconnect with purpose, and keep moving through ambiguity.

Organizational resilience: business model adaptation as strategic reflexivity

Organizational resilience manifested as the ability to restructure business models in real time, adapting offerings, operations, and strategies to rapidly shifting conditions. This area of analysis is closely linked to the previous one as cultural practices and routines enable informed and objective actions toward change and innovation. Across all cases, the most resilient startups were those that treated their business models as fluid, testable, and modular hypotheses rather than fixed plans.

During the pandemic, Batman transformed itself from a transactional travel provider into a purpose-driven community platform. With international travel halted, the team reoriented around identity and narrative. The founder shared:

> *We stopped asking, "Where are we sending people?" And started asking, "Who are we helping them become?" Our products became mirrors, tools for self-exploration. That insight reshaped everything, including the choice of local partners all over the world, the communication style, product development, and so forth.*

Through extensive experimentation (A/B testing landing pages, messages, price points), the company launched new offerings such as *Club Trips* and *Short Club*, centered on shared interests, guided experiences, and flexibility. The growth lead stated:

> *We didn't just redesign trips, we redesigned transformation. We mapped out identity arcs, not travel plans. The result? Our retention improved, even when travel was still partially restricted.*

Confronted with an unstable crypto market, Bane shifted from growth to retention. Its key insight was that better-educated users would be more loyal, stable, and profitable. The cofounder said:

We realized that our best users weren't the loudest but the most informed. So, we built a learning platform inside the investment experience—quizzes, modules, challenges. Engagement increased, and so did the average investment size. We shifted toward this different business model when we started facing very hard times.

The company also decided to develop a new product—crypto Exchange-Traded Fund (ETFs) for long-term planning—transforming it from a short-term speculative product into a financial service. This required a product shift along with an internal reorganization around cross-functional experimentation teams. This process was driven by a significant drop in user acquisition, marking a particularly challenging time for the startup.

Joker introduced an important structural shift. Once a listings directory for sports clubs, it rebuilt itself into an operational platform focused on services and a subscription model. The CEO explained:

Our clubs didn't need traffic. They needed automation. Billing, scheduling, check-ins. So, we became their infrastructure. We now power their operations, not just their marketing.

This shift was enabled by listening closely to clients and designing tools aligned with their survival, not just visibility. Once again, this ability to refocus and pivot stemmed from several routines and individual actions taken during a period of considerable difficulty. Courage was the driving force behind revisiting assumptions and experimenting with novel approaches.

Catwoman demonstrated agility in its operations by implementing machine learning for demand forecasting and route optimization while expanding offerings to new segments. The product manager noted:

We moved fast. From vegan meal bundles to corporate subscriptions, we had a new test running every week. Some failed fast, others scaled fast. But each taught us something.

Organizational resilience was evident in how Riddler, a B2B AI solution for pharmacy logistics, responded to operational breakdowns. After a software bug caused inventory failures that disrupted pharmaceutical deliveries, the team moved to an intensive repair mode. The founder recounted:

I personally visited seven pharmacies in three days. I sat behind the counter with them and watched how they worked. It was humbling. That's how we found the real issue; it wasn't just the bug. It was how our system communicated out-of-stock alerts. We rewrote that module in 72 h.

This is a case of hands-on immersion, with resilience enacted through proximity and customer co-repair, not just technical fixes. What followed was the rapid launch of a real-time dashboard for pharmacy inventory visibility, which became a new selling point. This highlights the need for adopting agile and lean methodologies for implementing new features and fixing bugs.

Nightwing responded to its retention crisis with radical iteration. The product team launched 12 onboarding experiments in six weeks, testing different flows, copy styles, and narrative hooks. One designer explained:

> *We stopped asking what features users wanted and started asking what kind of journey they wanted to go on. The best-performing onboarding didn't explain the app but told a story: where you are, where you could be, and how we'll get you there.*

This shift to story-centric product framing demonstrated how even design and User Experience (UX) choices could reflect deeper strategic resilience by using language and structure to rebuild user trust.

At Penguin, organizational resilience was a result of radical simplification. The company cut 80% of its codebase and focused on billing automation rather than expanding features to chase new users.

> *Our customers didn't need another dashboard. They needed peace of mind that the invoices were being paid. Thus, we rebuilt our core engine to be invisible, reliable, and boring.*

Across all cases, organizational resilience was not a one-time pivot but a continuous loop of sensing, experimenting, learning, and reconfiguring. It required a culture that could tolerate temporary inefficiency, ambiguity, and structural messiness in the pursuit of long-term fit.

Financial resilience: managing burn, monetizing assets, and timing capital

Our cases and results showed that financial resilience went beyond accessing more funds in the context of startups. It involved using existing capital intelligently, orchestrating resources creatively, and controlling narratives and timing in fundraising processes. The startups survived not because they were well-funded but because they were financially reflexive and able to reassign, renegotiate, and reimagine their financial structure.

Batman used the crisis to renegotiate contracts, convert fixed costs to variable ones, and implement community-driven acquisition. The founder recounted:

> *We couldn't pay for marketing, so we turned to our users. We launched a referral program: bring a friend and earn credits. Then, we turned the program into content. We asked users to share their travel stories, which gave us reach without using ads. Our customer acquisition cost dropped by 20%.*

Batman also bartered with hotels and local partners, exchanging exposure and feedback for discounted services, creating a new value exchange system.

Bane deliberately delayed its Series B, choosing to cut burn and refine metrics before reentering the market. The founder reflected:

We waited. Everyone else raised fast. We raised later, with better terms. That taught us that time is capital too. When you control time, you control valuation.

Joker monetized through low-cost conversion funnels. It offered free trials, tracked user behaviour, and built onboarding paths that nudged upgrades at key points. The CEO explained:

We didn't upsell with features. We upsold with understanding. We showed clubs what they could automate and what they were missing. When they saw the numbers, they converted.

This strategy stabilized cash flow without requiring capital infusion.

Furthermore, Catwoman transformed excess delivery capacity into a service.

Our rider fleet was underused midweek. Therefore, we offered delivery-as-a-service to small brands. It didn't make us rich, but it paid for itself, and that bought us three more months of experimentation.

Financial resilience was not a matter of resource abundance but of resource fluidity, which can be defined as the ability to reallocate quickly and repurpose or recombine existing assets into new value-generating configurations. Instead of passively enduring periods of low demand, Catwoman actively converted underutilized capacity into an interim revenue stream, demonstrating agility in cost management and buying time for further innovation. This fluidity allowed the company to remain operational without additional external capital.

From a financial perspective, Riddler faced a performance decrease just before the crisis. It had closed a small convertible note two weeks before the system failure. The founder noted:

It was pure luck and good timing. If the bug had hit before we signed, I doubt they would have invested. But we raised on vision, not on performance. That's something we learned—to raise when your narrative is strong, not when your metrics are weak.

This quote illustrated a key lesson in financial resilience—the importance of narrative control and timing as funding levers.

Nightwing operated without external funding and, thus, had to embrace extreme frugality. The founder shared:

Every euro we spent had to return something within a month. We ran ads only on TikTok, where Cost Per Mille (CPMs) were low and engagement was high. We even filmed the videos ourselves. The constraints made us smarter.

This constraint-driven creativity mindset became Nightwing's financial operating system, prioritizing lifetime value to customer acquisition cost ratios and weekly burn adjustments.

Penguin adopted a strategy of pricing restructuring and liquidity insulation. It introduced annual billing plans with flexible pause options.

> *We gave customers a way to "hibernate" their subscription without churning. It reduced churn, protected cash flow, and created goodwill. It was a small feature with a big impact.*

Financial resilience emerged through behavioral pricing psychology, offering users perceived flexibility while increasing predictability for the company.

Table 2 shows some examples of the resilience tactics.

Discussion of findings

This study offers an integrated perspective on how early-stage startups enact resilience through the dynamic interplay of psychological, organizational, and financial mechanisms. We did not treat resilience as a static trait but framed it as a recursive process built through iterative interpretation, resource orchestration, and shared sensemaking. Accordingly, our findings align with prior research on organizational resilience (Lengnick-Hall et al., 2011; Duchek, 2020) that conceptualizes resilience as a dynamic capability. Unlike earlier studies on established firms and stable routines, our early-stage evidence points to transient, performative micro-foundations of emotional rituals, modular experimentation, and financial pacing.

Each of the three dimensions supports and enables the others, forming a recursive system of coping, learning, and reinvention, which aligns with the view of resilience as an emergent, situated process (Lengnick-Hall et al., 2011; Korber & McNaughton, 2018) rather than a fixed organizational trait.

Psychological resilience emerged as a foundational condition for adaptation. Startup founders navigated emotional turbulence through emotional presence, vulnerability, and ritualized communication, not through denial or control. These behaviors provided cognitive and emotional scaffolding during volatile periods. Batman's daily calls, Bane's reframing, and Penguin's debriefs acted as stabilizing anchors that restored attention and emotional control—enactments of order under ambiguity (Weick, 1995). Based on these observations, we introduced two concepts. First, cognitive grounding is the practice of reconnecting with core users and purposes to regain mental coherence and clarity of direction. This mirrors Weick's (1993) "sensemaking in crisis," with actors rebuilding attention through simple, repeatable acts. In our cases, the teams shifted from metric-driven decisions to user-centered listening. Second, affective anchoring is the deployment of rituals or symbolic gestures to stabilize team emotions and foster shared understanding under pressure. These anchors did not provide strategic answers but restored the emotional capacity to absorb complexity, such as the stabilizing function of routines discussed by Feldman and Pentland (2003). Both mechanisms underscore the relational and narrative dimensions of resilience (Maitlis & Sonenshein, 2010). Finally, while prior research has highlighted individual traits like optimism and self-regulation (Masten, 2001; Chadwick & Raver, 2020), our evidence showed that psychological resilience in startups was often socially distributed and emerged from collective rituals and symbolic framing.

Table 2 Examples of resilience

Startup	Psychological Resilience	Organizational Resilience	Financial Resilience
Batman	Daily calls to maintain team presence and identity; reframing loss as transformation.	Pivot to identity-based travel (Club Trips, Short Club); A/B testing on packages.	Referral-based growth, bartering with suppliers, renegotiating costs.
Bane	Emotional containment under crypto market stress; 'We are the lighthouse' metaphor.	Shift from user acquisition to education-based retention; embedded learning platform.	Delayed Series B to preserve valuation; controlled burn rate, value-driven fundraising.
Joker	Empathic calls with club managers; support reframed as emotional infrastructure.	Pivot from directory to SaaS; tools for automation, CRM, billing.	Freemium conversion funnel; upselling via user understanding, not features.
Catwoman	Post-growth disorientation managed through 'Friday Reflections' ritual.	ML-powered demand forecasting; rapid experimentation with segment-based offers.	Bundle pre-payment models; logistics-as-a-service monetization.
Nightwing	Founder joined support calls to reconnect with user pain points.	12 onboarding experiments in 6 weeks; story-based product framing.	Frugal TikTok-based marketing; self-filmed content to stretch budget.
Penguin	CTO's reframing of churn: focus on those who stayed.	Core simplification to invoicing tools; codebase reduction.	Annual plan + pause feature; protected liquidity and reduced churn.
Riddler	Founder's personal visits to pharmacies to absorb frontline frustration.	Real-time inventory dashboard developed post-crisis; immersion-based redesign.	Closed note pre-crisis based on narrative; raised before metrics collapse.

This nuance contributes to the literature by shifting the focus from personality traits to team-level affective practices.

This psychological grounding enabled startups to access and sustain organizational resilience, defined here as reflexive adaptiveness and not robustness This adaptive capability does not depend on single "pivots" but on cycles of micro-adjustments, iterative learning, and modular redesign. This reflects the spirit of the lean startup approach (Ries, 2011) and extends it by showing how emotional safety and narrative clarity are prerequisites for experimentation. We identified three interconnected capabilities that foster organizational and strategic resilience. First, experimental respon-

siveness captured how founders used structured hypotheses and low-cost tests to navigate complexity while minimizing cognitive overload. Second, modular recomposition referred to the ability to reconfigure business model elements, such as pricing, delivery, or UX, without compromising coherence and vision. Third, sense-giving, as theorized by Gioia and Chittipeddi (1991), was also pervasive, with founders actively redefining the firm's identity in response to environmental changes, helping teams interpret shifts not as threats but as trajectories of growth. Accordingly, organizational resilience was not only a response to shock but an ongoing project of redefinition and was sustained by interaction with users and internal teams. Therefore, as founders and teams learned how to cope with difficulties, they also developed the capacity to pivot and transform their activities, products, services, and business models.

Financial resilience was more about strategic timing, liquidity control, and narrative leverage than capital availability. When addressing limited resources, startups displayed financial reflexivity, modulating burn, adjusting monetization logic, and orchestrating fundraising in alignment with internal conditions and market perception. These practices were active tools of temporal strategy and shaped the pace of experimentation and organizational change. In this regard, cash flow tactics involved the careful alignment of income and expenditure to preserve flexibility without sacrificing focus. Resource reconfiguration, in line with Teece's (2007) dynamic capabilities framework, was evident when idle assets, like rider capacity at Catwoman, were repurposed into short-term revenue models. Finally, temporal elasticity allowed founders to compress or stretch strategic time horizons by controlling their capital rhythm, which echoed the insights of Bhamra et al. (2011) on time-based resilience. In conclusion, unlike most studies that treat financial resilience as a function of access to capital or planning (e.g., Sreenivasan & Suresh, 2023), our findings highlighted temporal elasticity and narrative control as strategic levers of financial resilience. These elements, which are largely overlooked in the literature, demonstrate how founders can shape investor perceptions and timing as part of their resilience strategy.

These three crucial domains did not operate in silos. We observed a recursive, mutually reinforcing loop. Psychological rituals created the stability needed for strategic experimentation; experimentation required time and tolerance for failure, sustained by financial buffering; and the financial strategy depended on collective clarity and team alignment. At Batman, emotional continuity fostered identity-based business model innovation, which led to creative acquisition tactics that reduced customer acquisition cost and preserved liquidity. At Joker, empathic outreach unlocked a new SaaS model, which necessitated pricing innovation and cash flow adjustments. These cases demonstrated that startup resilience was not linear but layered, iterative, and deeply interwoven with narrative and emotional labor.

Figure 1 presents the concept of startup resilience as a recursive recomposition process built on three interdependent psychological, organizational or strategic, and financial layers. Each layer comprises analytically distinct micro-mechanisms that enable adaptation under adversity through feedback loops.

As previously discussed, psychological resilience is the capacity to stabilize attention and address emotions to allow the team to continue functioning under pressure. It is based on three micro-mechanisms: (a) cognitive grounding (sensemaking routines that recenter purpose, users, and facts to reduce noise and reestablish clarity), (b)

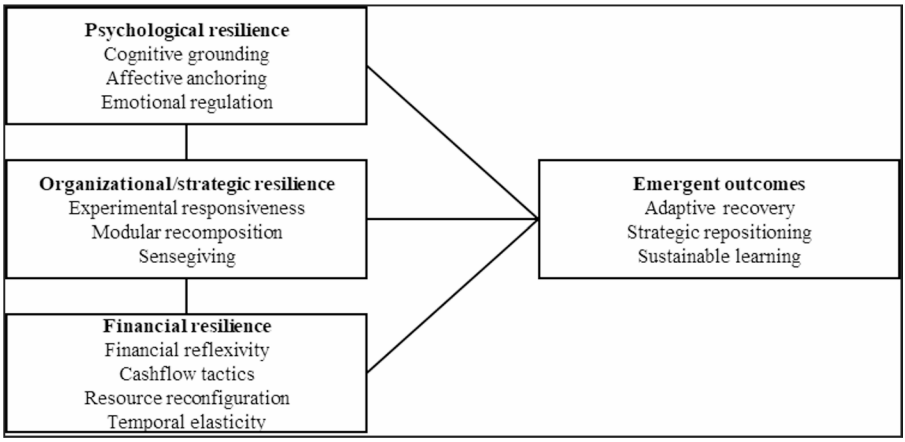

Fig. 1 Resilience as a recursive recomposition process. (Source: Authors)

affective anchoring (shared rituals and symbols that stabilize the team's emotional climate and sustain collective agency), and (c) emotion regulation (individual strategies that maintain arousal within functional bounds and preserve decision quality). Organizational or strategic resilience is the ability to translate signals into coordinated change while preserving strategic coherence. It operates through experimental responsiveness (tight build–measure–learn cycles triggered by environmental cues), modular recomposition (reassembling business-model and process modules via modular interfaces so that parts can change without rebuilding the whole), and sensegiving (leadership framing that aligns internal actors and secures stakeholder buy-in for change).

Finally, financial resilience buffers against shocks and sustains adaptive capacity under resource constraints. It involves financial reflexivity (continuous monitoring and adaptive pacing of cash, burn, and runway), cash flow tactics (short-term working-capital maneuvers that stabilize liquidity), resource reconfiguration (reallocation of financial resources and restructuring of commitments), and temporal elasticity (timing flexibility that intertemporally shifts cash flows and funding events to extend runway and preserve strategic leverage).

These mechanisms interact recursively: psychological stability enables rapid experimentation; experimentation informs financial pacing; and financial elasticity creates time and option value for further cognitive, organizational, and strategic adjustment. Ultimately, this yields emergent outcomes such as adaptive recovery, motivational stabilization, strategic repositioning, business-model redefinition, sustainable learning, and the long-term integration of resilient routines and resource logic.

Through recursive interaction, these micro-mechanisms generate three higher-order outcomes: adaptive recovery (restoration of viable functioning at a new, more resilient equilibrium instead of a mere return to the status quo ante), strategic repositioning (deliberate reorientation of target segments, value proposition, go-to-market, or revenue logic to fit altered conditions), and sustainable learning (an institutional-

ized learning cadence—routines, artifacts, and metrics—that persistently converts experimentation into improved future adaptation).

Table 3 summarizes the study's construct architecture.

Implications

Theoretical implications

On a theoretical level, our study offers the following implications. First, the study contributes to the literature on resilience in startups (Silva et al., 2023; Sreenivasan et al., 2023). It reconceptualizes startup resilience not as a fixed organizational trait or reactive capacity but a recursive recompositional process enacted through iterative adaptation, emotional grounding, and financial maneuvering. This perspective challenges linear models of "bounce-back" or "absorptive" resilience and emphasizes the dynamic, nonlinear nature of entrepreneurial adaptation, which can also introduce a substantial strategic change. Therefore, we contribute to the organizational resilience literature (Lengnick-Hall et al., 2011; Duchek, 2020) by showing that, in early-stage ventures, resilience is enacted not through stable structures but through the continuous redefinition of business models, routines, and cognitive frames. Second, the study adds to the growing literature on the micro-foundations of entrepreneurial resilience (Ayala & Manzano, 2014; Chadwick & Raver, 2020) by identifying previously underexplored affective mechanisms such as cognitive grounding, affective anchoring, and emotional regulation rituals. These mechanisms reveal that psychological resilience in startups concerns not just the founder's personal traits but a collectively sustained emotional and relational architecture that can be fostered through team routines, metaphors, and interactional patterns. Third, by building on the dynamic capabilities view, we reveal how startups redesign their business models through modular recomposition, experimental responsiveness, and sense-giving (Teece et al., 1997; O'Reilly & Tushman, 2013). We demonstrate that dynamic capabilities in startups often emerge under conditions of structural incompleteness and emotional pressure that require agility in sensing, seizing, and reconfiguring internal logics. Moreover, the findings introduce the concept of resourcefulness-as-capability, in which the ability to creatively reallocate or reframe existing resources becomes a core enabler of resilience. Finally, this study identifies financial reflexivity, cash flow tactics, and temporal elasticity as key levers through which startups mediate between survival and strategic experimentation. These mechanisms illustrate that financial resilience is not just about capital availability but involves the cognitive and narrative control of financial time.

Practical implications

These findings offer several actionable insights for founders and managers seeking to build resilience in their startups. Resilient startups combine emotional grounding, structured experimentation, and financial resourcefulness. These practices, observed across diverse cases, underscore a set of practical routines, tactics, and postures that can be implemented even with limited resources. First, psychological resilience is best cultivated through shared rituals, metaphors, and emotional transparency, not

Table 3 Concepts, definitions and descriptions

Layer	Concept	Concise definition	Boundaries/operational indicators
Psychological resilience	**Psychological resilience (macro)**	Capacity of founders/teams to stabilize attention and affect to preserve functioning and decision quality under pressure.	Distinct from "morale" or performance; focuses on cognitive–affective stability that enables action. *Indicators*: brief emotional check-ins, stable focus time, reduced affective volatility during shocks.
Psychological	**Cognitive grounding**	Sensemaking routines that re-center purpose, users, and facts to reduce noise and regain clarity.	*Examples*: user check-ins; revisiting North-Star metrics; restating mission/problem.
Psychological	**Affective anchoring (rituals)**	Shared rituals/symbols that stabilize the team's emotional climate and sustain collective agency.	*Examples*: scripted retros, peer-recognition moments, narrative storytelling.
Psychological	**Emotion regulation**	Individual strategies—primarily cognitive reappraisal—that keep arousal within functional bounds.	*Examples*: structured breaks, breathing protocols, reframing exercises.
Organizational/Strategic resilience	**Organizational/Strategic resilience (macro)**	Ability to translate signals into coordinated change while preserving strategic coherence and continuity.	Distinct from psychological (climate) and financial (liquidity/structure). *Outputs*: BM iterations, process realignment.
Organizational/Strategic	**Experimental responsiveness**	Tight build–measure–learn cycles triggered by environmental cues to reduce uncertainty and speed decisions.	*Examples*: rapid A/B tests, targeted MVPs, feature flags, explicit kill/keep rules.
Organizational/Strategic	**Modular recomposition**	Reassembling business-model/process modules via interfaces so parts can change without rebuilding the whole.	*Examples*: channel switch, partner substitution, pricing module redesign.
Organizational/Strategic	**Sensegiving**	Leadership framing/narrative that aligns internal actors and secures stakeholder buy-in for change.	*Examples*: alignment town-halls, OKR reframing, concise "why–what–how" decks.
Financial resilience	**Financial resilience (macro)**	Shock-buffering and adaptive capacity under resource constraints through monitoring, allocation, and timing.	Distinct from organizational reconfiguration; concerns liquidity, capital structure, and intertemporal choices.
Financial	**Financial reflexivity**	Continuous monitoring and **adaptive pacing** of cash, burn, and runway that links financial signals to action.	*Examples*: dynamic spend thresholds; pricing tweaks tied to runway; monthly runway reviews.

Table 3 (continued)

Layer	Concept	Concise definition	Boundaries/operational indicators
Financial	**Cash-flow tactics**	Short-term working-capital maneuvers that stabilize liquidity.	*Examples*: advance payments, early-pay discounts, stretching payables, throttling variable costs.
Financial	**Resource reconfiguration**	Reallocation of financial resources and restructuring of commitments.	*Examples*: budget shifts, asset redeployment, contract renegotiation, debt/equity mix adjustments.
Financial	**Temporal elasticity**	Timing flexibility that intertemporally shifts cash flows and funding events to extend runway and preserve leverage.	*Examples*: phased hiring, deferred capex, fundraising after traction milestones.
Emergent outcomes	**Emergent outcomes (macro)**	System-level patterns produced by recursive interaction of the three layers.	Higher-order, cross-case effects rather than single actions.
Emergent outcomes	**Adaptive recovery**	Restoration of viable functioning at a *new*, *more resilient equilibrium* after a shock.	*Indicators*: time-to-recovery of MRR/churn/uptime; stabilized cash/runway with adjusted processes/offers.
Emergent outcomes	**Strategic repositioning**	Deliberate re-orientation of target segments, value proposition, go-to-market, or revenue logic.	*Indicators*: new ICP/segment focus; revised pricing tiers; channel/partner switch; updated OKRs/roadmap.
Emergent outcomes	**Sustainable learning**	Institutionalized learning cadence (routines, artifacts, metrics) that continuously converts experimentation into improved future adaptation.	*Indicators*: documented experiment backlog; regular review cadence; instrumented North-Star/guardrails; shorter cycle time; rising experiment win-rate.

through motivational speeches or individual heroism. Founders can create stability by institutionalizing simple and consistent routines, such as daily check-ins and weekly reflections, that help teams process uncertainty collectively. Additionally, the use of clear metaphors can provide teams with cognitive anchors that help reframe chaos and restore composure. During moments of strategic doubt or market pressure, returning to direct user interaction, rather than overreliance on metrics, can help founders reconnect with the core problem they are solving and ensure clarity of direction. Second, organizational resilience concerns modular experimentation and meaning-making instead of bold, reactive pivots. Founders are encouraged to deconstruct their business model into testable components and run small, focused experiments before implementing major strategic changes. This approach allows for targeted adaptation without destabilizing overall coherence. Third, financial resilience is less about resource abundance and more about timing, creativity, and liquid-

ity control. Founders should actively engage in cash flow tactics, specifically the careful management of the timing of payments, revenue inflows, and cost structures to extend runway without additional funding. Furthermore, founders should delay external fundraising unless their strategic narrative is strong and supported by clear learning outcomes. Raising capital under weak circumstances may limit strategic options and valuation leverage.

Conclusions

This study examined how resilience is enacted within startups by analyzing its underlying psychological, organizational, and financial mechanisms. Drawing on multiple case studies of seven early-stage ventures across various industries, we demonstrated that resilience in startups was not a fixed capability but a recursive, processual phenomenon. Our analysis revealed how founders and teams co-constructed resilience through emotional grounding, strategic responsiveness, and financial discipline, thereby enabling adaptation and continuity under uncertainty. Psychological resilience emerged through mechanisms such as cognitive grounding and affective anchoring. Additionally, founders created shared rituals and reframed adversity in ways that stabilized their teams and restored clarity of purpose. Organizational resilience was enacted via iterative experimentation and the modular recomposition of business models and supported by sense-giving narratives that facilitated coordinated action. Financial resilience manifested through resource fluidity, cash flow tactics, and temporal elasticity, which enabled startups to align financial behavior with strategic intent and maintain operational runway.

As with all qualitative studies, our research has some limitations that unlock avenues for future investigations. First, our sample comprised seven startups, primarily operating in European or Western markets. While this allowed for an in-depth case analysis, it limited the generalizability of the findings to other geographical or institutional contexts, particularly in emerging economies. Future research should replicate or extend our framework across different regions or socioeconomic conditions to explore contextual variations in the enactment of resilience. Second, although the cases represented several industries and maturity levels, they were platform-based or digitally enabled ventures. This might have influenced the types of resilience mechanisms observed, particularly with respect to agility, resourcefulness, and financial fluidity. Research on non-digital, product-based, or service-heavy startups could reveal different pathways to resilience. Third, our reliance on retrospective interviews and secondary data might have introduced recall or desirability biases, despite attempts to mitigate this risk through triangulation and member checking. Longitudinal or ethnographic designs could help capture resilience as it unfolds in real time, offering a richer processual view. Finally, while our framework was grounded in empirical evidence, it did not assess the causal relationships among the identified mechanisms. Future studies should adopt mixed-methods or longitudinal quantitative approaches to validate and refine the recursive relationships between psychological, organizational, and financial dimensions. Linking resilience mechanisms to measurable performance outcomes could further enhance the practical value of this research stream.

Appendix 1

Table 4 Interview questions

Crisis Context	Can you describe a particularly difficult or critical moment your startup experienced?	Williams and Shepherd (2016)
Crisis Context	What were the main impacts of that situation—on your operations, people, and market positioning?	Williams and Shepherd (2016)
Crisis Context	Looking back, would you characterize that crisis as internal or external? Sudden or gradual?	Williams and Shepherd (2016)
Crisis Context	At what point did you realize that your response needed to go beyond "business as usual"?	Williams and Shepherd (2016)
Psychological Resilience	How did your team react during those difficult times? What helped (or hindered) the human side of the response?	Chadwick and Raver (2020)
Psychological Resilience	As a founder/manager, how did you personally cope with uncertainty and stress?	Chadwick and Raver (2020)
Psychological Resilience	To what extent do you believe your personal traits (e.g., perseverance, optimism, self-regulation) made a difference?	Chadwick and Raver (2020)
Psychological Resilience	How important was trust, openness, and communication inside the team?	Chadwick and Raver (2020)
Psychological Resilience	Did you adopt any new practices to strengthen team motivation or cohesion under pressure?	Chadwick and Raver (2020)
Organizational Resilience	Did you change your business model, product offering, or strategic direction? How and why?	Mota et al. (2022)
Organizational Resilience	How were key decisions made during times of uncertainty? Who was involved?	Mota et al. (2022)
Organizational Resilience	Did you apply any particular methodologies (e.g., Lean Startup, agile sprints, growth loops) to stay adaptive?	Mota et al. (2022)
Organizational Resilience	What organizational routines or capacities were crucial to maintaining operational focus?	Mota et al. (2022)
Organizational Resilience	Did the crisis unlock unexpected opportunities for innovation or growth?	Mota et al. (2022)
Organizational Resilience	Was there a significant pivot? What conditions made that possible?	Mota et al. (2022)
Financial Resilience	How did you manage cash flow and financial resources during the crisis?	Sreenivasan and Suresh (2023)
Financial Resilience	Did you seek external financial support (investors, loans, grants)? What were the outcomes?	Sreenivasan and Suresh (2023)
Financial Resilience	How important was cost flexibility in helping your startup remain resilient?	Sreenivasan and Suresh (2023)
Financial Resilience	Did you have to renegotiate, downsize, or redesign any parts of the business to stay afloat?	Sreenivasan and Suresh (2023)
Financial Resilience	How did your relationship with external stakeholders affect your ability to bounce back?	Sreenivasan and Suresh (2023)
Lessons & Outlook	What are the most important lessons you've learned from navigating those difficult periods?	Duchek (2020)
Lessons & Outlook	Has your personal or team-level definition of "resilience" changed since those events?	Duchek (2020)
Lessons & Outlook	Have you introduced mechanisms to better prepare for future crises?	Duchek (2020)
Lessons & Outlook	What advice would you offer to other founders going through similar turbulence?	Duchek (2020)
Lessons & Outlook	Looking ahead, what would you prioritize to strengthen your startup's resilience further?	Duchek (2020)

References

Ayala, J. C., & Manzano, G. (2014). The resilience of the entrepreneur. Influence on the success of the business. A longitudinal analysis. *Journal of Economic Psychology, 42*, 126–135.

Bargoni, A., Santoro, G., Petruzzelli, A. M., & Ferraris, A. (2024a). Growth hacking: A critical review to clarify its meaning and guide its practical application. *Technological Forecasting and Social Change*, *200*, 123111.

Bargoni, A., Smrčka, L., Santoro, G., & Ferraris, A. (2024b). Highway to hell or paradise city? Exploring the role of growth hacking in learning from innovation failure. *Technovation*, *131*, 102945.

Bhamra, R., Dani, S., & Burnard, K. (2011). Resilience: The concept, a literature review and future directions. *International Journal of Production Research, 49*(18), 5375–5393. https://doi.org/10.1080/00207543.2011.563826

Blank, S. (2018). Startups are not a smaller version of a large company. *The gurubook* (pp. 130–133). Productivity.

Bohnsack, R., & Liesner, M. M. (2019). What the hack? A growth hacking taxonomy and practical applications for firms. *Business Horizons, 62*(6), 799–818.

Branca, E., Intenza, M., & Doni, F. (2025). Startup entrepreneurs' personality traits and resilience: Unveiling the interplay of prior experience. *International Entrepreneurship and Management Journal*, *21*(1), 2.

Brykman, K. M., & King, D. D. (2021). A resource model of team resilience capacity and learning. *Group & Organization Management, 46*(4), 737–772.

Bullough, A., & Renko, M. (2013). Entrepreneurial resilience during challenging times. *Business Horizons*, *56*(3), 343–350.

Castelló-Sirvent, F., Peris-Ortiz, M., Llopis-Amorós, M., & Pinazo-Dallenbach, P. (2024). How does the COVID-19 economic crisis impact resilience? A configurational analysis of the spinoffs. *International Entrepreneurship and Management Journal*, *20*(3), 1823–1848.

Cavallo, A., Cosenz, F., & Noto, G. (2024). Business model scaling and growth hacking in digital entrepreneurship. *Journal of Small Business Management*, *62*(4), 2058–2085.

Chadwick, I. C., & Raver, J. L. (2020). Psychological resilience and its downstream effects for business survival in nascent entrepreneurship. *Entrepreneurship Theory and Practice, 44*(2), 233–255.

Cloutier, C., & Ravasi, D. (2021). Using tables to enhance trustworthiness in qualitative research. *Strategic Organization, 19*(1), 113–133. https://doi.org/10.1177/1476127020979329

Conz, E., & Magnani, G. (2020). A dynamic perspective on the resilience of firms: A systematic literature review and a framework for future research. *European Management Journal*, *38*(3), 400–412.

Corvello, V., Verteramo, S., Nocella, I., & Ammirato, S. (2023). Thrive during a crisis: The role of digital technologies in fostering antifragility in small and medium-sized enterprises. *Journal of Ambient Intelligence and Humanized Computing*, *14*(11), 14681–14693.

Corvello, V., Felicetti, A. M., Pini, B., & Bigliardi, B. (2024). International entrepreneurship as a dynamic capability in new ventures: Defining the concept and identifying its antecedents and outcomes through a systematic literature review. *Journal of International Entrepreneurship*(3). https://doi.org/10.1007/s10843-024-00368-3

Cristofaro, M., Giardino, P. L., & Barboni, L. (2025). Growth hacking: A scientific approach for data-driven decision making. *Journal of Business Research, 186*, Article 115030.

Cuéllar-Fernández, B., Fuertes-Callén, Y., & Serrano-Magdalena, A. (2024). Factors behind the resilience of rural startups. *Technological Forecasting and Social Change, 206*, Article 123521.

Doz, Y. L., & Kosonen, M. (2010). Embedding strategic agility: A leadership agenda for accelerating business model renewal. *Long Range Planning, 43*(2–3), 370–382.

Duchek, S. (2018). Entrepreneurial resilience: A biographical analysis of successful entrepreneurs. *International Entrepreneurship and Management Journal*, *14*(2), 429–455.

Duchek, S. (2020). Organizational resilience: A capability-based conceptualization. *Business Research*, *13*(1), 215–246.

Eisenhardt, K. M. (1989). Building theories from case study research. *Academy of Management Review, 14*(4), 532–550.

Eisenhardt, K. M., & Graebner, M. E. (2007). Theory building from cases: Opportunities and challenges. *Academy of Management Journal, 50*(1), 25–32. https://doi.org/10.5465/amj.2007.24160888

Feldman, M. S., & Pentland, B. T. (2003). Reconceptualizing organizational routines as a source of flexibility and change. *Administrative Science Quarterly, 48*(1), 94–118. https://doi.org/10.2307/3556620

Flechas Chaparro, X. A., & de Vasconcelos Gomes, L. A. (2021). Pivot decisions in startups: A systematic literature review. *International Journal of Entrepreneurial Behavior & Research, 27*(4), 884–910.

Fuertes-Callén, Y., Cuellar-Fernández, B., & Serrano-Cinca, C. (2022). Predicting startup survival using first years financial statements. *Journal of Small Business Management*, *60*(6), 1314–1350.

Gioia, D. A., & Chittipeddi, K. (1991). Sensemaking and sensegiving in strategic change initiation. *Strategic Management Journal, 12*(6), 433–448. https://doi.org/10.1002/smj.4250120604

Grego, M., Magnani, G., & Denicolai, S. (2024). Transform to adapt or resilient by design? How organizations can foster resilience through business model transformation. *Journal of Business Research, 171*, Article 114359.

Haase, A., & Eberl, P. (2019). The challenges of routinizing for building resilient startups. *Journal of Small Business Management, 57*, 579–597.

Hedner, T., Abouzeedan, A., & Klofsten, M. (2011). Entrepreneurial resilience. *Annals of Innovation & Entrepreneurship, 2*(1), Article 7986.

Hillmann, J., & Guenther, E. (2021). Organizational resilience: A valuable construct for management research? *International Journal of Management Reviews*, *23*(1), 7–44.

Holling, C. S. (1973). Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*

Kim, J. R. (2023). Jumping the hurdles in making new venture: The mediating role of marketing self-efficacy in start-up intention. *Global Business & Finance Review (GBFR), 28*(4), 51–61.

Korber, S., & McNaughton, R. B. (2018). Resilience and entrepreneurship: A systematic literature review. *International Journal of Entrepreneurial Behavior & Research, 24*(7), 1129–1154. https://doi.org/10.1108/IJEBR-10-2016-0356

Kuckertz, A., Brändle, L., Gaudig, A., Hinderer, S., Reyes, C. A. M., Prochotta, A., & Berger, E. S. (2020). Startups in times of crisis–A rapid response to the COVID-19 pandemic. *Journal of Business Venturing Insights*, *13*, e00169.

Lengnick-Hall, C. A., Beck, T. E., & Lengnick-Hall, M. L. (2011). Developing a capacity for organizational resilience through strategic human resource management. *Human Resource Management Review, 21*(3), 243–255.

Lopez, E., Flecha-Ortiz, J. A., Santos-Corrada, M., & Dones, V. (2024). The role of organizational resilience in SME service innovation and value cocreation. *Journal of Services Marketing, 38*(4), 443–459.

Luthans, F., Avolio, B. J., Avey, J. B., & Norman, S. M. (2007). Positive psychological capital: Measurement and relationship with performance and satisfaction. Personnel psychology, 60(3), 541–572.

Macca, L. S., Santoro, G., Jabeen, F., & Gavurova, B. (2025). Fuelling growth: A qualitative study on the benefits and challenges of growth hacking for micro, small and medium enterprises. *International Journal of Entrepreneurial Behavior & Research*. https://doi.org/10.1108/ijebr-05-2024-0520

Maitlis, S., & Sonenshein, S. (2010). Sensemaking in crisis and change: Inspiration and insights from Weick (1988). *Journal of Management Studies, 47*(3), 551–580. https://doi.org/10.1111/j.1467-6486.2010.00908.x

Masten, A. S. (2001). Ordinary magic: Resilience processes in development. *American Psychologist, 56*(3), 227.

Morris, M. H., Kuratko, D. F., Audretsch, D. B., & Santos, S. (2020). Overcoming the liability of poorness: Disadvantage, fragility, and the poverty entrepreneur. *Small Business Economics,* (1), 15. https://doi.org/10.1007/s11187-020-00409-w

Morris, J., Morris, W., & Bowen, R. (2022). Implications of the digital divide on rural SME resilience. *Journal of Rural Studies*, *89*, 369–377.

Mota, R. D. O., Bueno, A., Gonella, J. D. S. L., Ganga, G. M. D., Godinho Filho, M., & Latan, H. (2022). The effects of the COVID-19 crisis on startups' performance: The role of resilience. *Management Decision, 60*(12), 3388–3415.

O'Reilly III, C. A., & Tushman, M. L. (2013). Organizational ambidexterity: Past, present, and future. *Academy of management Perspectives, 27*(4), 324–338.

Palmucci, D. N., Jabeen, F., & Santoro, G. (2024). Customer insights in the digital era: A mixed-method approach to Understanding e-service interactions. *EuroMed Journal of Business*.

Rehman, S. U., Jabeen, F., Shahzad, K., Riaz, A., & Bhatti, A. (2025). Industry 4.0 technologies and international performance of SMEs: Mediated-moderated perspectives. *International Entrepreneurship and Management Journal, 21*(1), 25.

Ries, E. (2011). *The lean startup: How today's entrepreneurs use continuous innovation to create radically successful businesses.* Crown Business

Roffia, P., & Dabić, M. (2024). The role of management control and integrated information systems for the resilience of SMEs. *Review of Managerial Science*, *18*(5), 1353–1375.

Roundy, P. T., Brockman, B. K., & Bradshaw, M. (2017). The resilience of entrepreneurial ecosystems. *Journal of Business Venturing Insights, 8*, 99–104.

Sage. Weick, K. E. (1993). The collapse of sensemaking in organizations: The Mann Gulch disaster. *Administrative Science Quarterly, 38*(4), 628–652. https://doi.org/10.2307/2393339

Sanasi, S. (2023). Entrepreneurial experimentation in business model dynamics: Current understanding and future opportunities. *International Entrepreneurship and Management Journal, 19*(2), 805–836.

Santoro, G., Bertoldi, B., Giachino, C., & Candelo, E. (2020). Exploring the relationship between entrepreneurial resilience and success: The moderating role of stakeholders' engagement. *Journal of Business Research*, *119*, 142–150.

Santoro, G., Messeni-Petruzzelli, A., & Del Giudice, M. (2021). Searching for resilience: The impact of employee-level and entrepreneur-level resilience on firm performance in small family firms. *Small Business Economics, 57*(1), 455–471.

Santos, S. C., Liguori, E. W., & Garvey, E. (2023). How digitalization reinvented entrepreneurial resilience during COVID-19. *Technological Forecasting and Social Change, 189*, Article 122398.

Silva, D. S., Ghezzi, A., Aguiar, R. B. D., Cortimiglia, M. N., & ten Ca, C. S. (2020). Lean startup, agile methodologies and customer development for business model innovation: A systematic review and research agenda. *International Journal of Entrepreneurial Behavior & Research, 26*(4), 595–628.

Silva, E., Beirão, G., & Torres, A. (2023). How startups and entrepreneurs survived in times of pandemic crisis: Implications and challenges for managing uncertainty. *Journal of Small Business Strategy*, *33*(1), 84–97.

Sommer, S. C., Loch, C. H., & Dong, J. (2009). Managing complexity and unforeseeable uncertainty in startup companies: An empirical study. *Organization Science, 20*(1), 118–133.

Sreenivasan, A., & Suresh, M. (2023). Readiness of financial resilience in start-ups. *Journal of Safety Science and Resilience*, *4*(3), 241–252.

Sreenivasan, A., Suresh, M., & Tuesta Panduro, J. A. (2023). Modelling the resilience of start-ups during COVID-19 pandemic. *Benchmarking : An International Journal, 30*(6), 2085–2109.

Strauss, A. L., & Corbin, J. (1998). Basics of qualitative research: Techniques and procedures for developing grounded theory (2nd ed.). *Sage*.

Taghavy, A., Hazari, N., & Hooshmand Chaijani, M. (2024). Assessing the impact of dynamic capabilities, resilience and strategic alignment on startup competitiveness in Iran. *Asia-Pacific Journal of Business Administration*.

Tedeschi, R. G., & Calhoun, L. G. (2004). Posttraumatic growth: conceptual foundations and empirical evidence. *Psychological inquiry, 15*(1), 1–18.

Teece, D. J. (2007). Explicating dynamic capabilities: the nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal, 28*(13), 1319–1350.

Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal, 18*(7), 509–533.

Teece, D. J., Peteraf, M., & Leih, S. (2016). Dynamic capabilities and organizational agility: Risk, uncertainty, and strategy in the innovation economy. *California Management Review*, *58*(4), 13–35.

Vogus, T. J., & Sutcliffe, K. M. (2007, October). Organizational resilience: Towards a theory and research agenda. In *2007 IEEE international conference on systems, man and cybernetics* (pp. 3418–3422). IEEE.

Weick, K. E. (1995). *Sensemaking in Organizations*. Sage.

Williams, T. A., & Shepherd, D. A. (2016). Building resilience or providing sustenance: Different paths of emergent ventures in the aftermath of the Haiti earthquake. *Academy of Management Journal, 59*(6), 2069–2102.

Yin, R. K. (2018). Case study research and applications: Design and methods (6th ed.). *SAGE*.

Zahra, S. A. (2021). The resource-based view, resourcefulness, and resource management in startup firms: A proposed research agenda. *Journal of Management, 47*(7), 1841–1860.

Zott, C., & Amit, R. (2008). The fit between product market strategy and business model: Implications for firm performance. *Strategic Management Journal*, *29*(1), 1–26.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Fauzia Jabeen[1] · Gabriele Santoro[2,3] · Tomas Kliestik[4] · Dorra Yahiaoui[5]

✉ Fauzia Jabeen
fauzia.jabeen@uaeu.ac.ae

Gabriele Santoro
gabriele.santoro@unito.it

Dorra Yahiaoui
dorra.yahiaoui@kedgebs.com

1 Department of Marketing and Entrepreneurship, College of Business and Economics, United Arab Emirates University, Al Ain, United Arab Emirates

2 Department of Management, University of Turin, Turin, Italy

3 Gnosis: Mediterranean Institute for Management Science, University of Nicosia School of Business, Nicosia, Cyprus

4 University of Zilina, Zilina, Slovakia

5 Kedge Business School, Talence, France